DSC Capstone LaTeX Template

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Welcome to the Capstone LaTeX template! To use this template:

- Create an Overleaf account (it's free).
- Click the "Menu" button in the top left, then click "Copy Project."
- To share the project with your group members, click "Share" in the top menu bar, turn on link sharing, and send the edit link to your group members.

The lines you see in blue are instructions from us; **remove them** or comment them out with `%` as you actually edit this template. **Your actual submissions should not have any blue in them!**

This template mostly exists for style purposes. **Not all headings provided in the template will actually be useful for you; you can add or remove headings as you please, and the headings you end up using are up to you and your mentor.** Look to the "Anatomy of a Scientific Paper" section of Methodology Lesson 4 for guidance on which headings to use, and Methodology Lesson 7 for more resources on LaTeX.

Abstract

Replace the `lipsum[1]` in the source code with your abstract. Don't include an abstract with your Quarter 2 Proposal. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

For your Quarter 1 Project, you won't link a website, but will link your code. For your Quarter 2 Proposal, you won't link either of these. For your actual Quarter 2 Project, you'll link both of these.

Website: https://abc.github.io/
Code: https://github.com/abc

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1 Introduction

2 Methods

3 Results

4 Discussion

5 Conclusion

6 LaTeX Typesetting Examples

This is not a real section; it's just here to show examples of how to format various components. Remove it before submitting!

6.1 LaTeX Basics

Here's **bold** and *italicized* text. Here's `text_that_looks.like(code)`.

- Here's a regular bulleted list item.
- And another.

Here's a hyperlink. If you want to use a numbered list, you can experiment with:

1. This.
2. This.
3. And this.

Here's how you might include a snippet of actual code:

```
# If you want to use syntax highlighting, look into the minted package.
def f(x):
    return 2 * x + 3
```

Here's how you might format a single equation:

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

And a chain of equations:

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 &= \frac{1}{n}\sum_{i=1}^{n}(x_i^2-2x_i\bar{x}+\bar{x}^2)\\
&= \frac{1}{n}\sum_{i=1}^{n}x_i^2-\frac{2}{n}\bar{x}\sum_{i=1}^{n}x_i+\frac{\bar{x}^2}{n}\sum_{i=1}^{n}1\\
&= \frac{1}{n}\sum_{i=1}^{n}x_i^2-2\bar{x}^2+\bar{x}^2\\
&= \frac{1}{n}\sum_{i=1}^{n}x_i^2-\bar{x}^2
\end{aligned}$$

6.2 Figure Examples

Here are some example figures. Figure 1 presents a scatter plot.

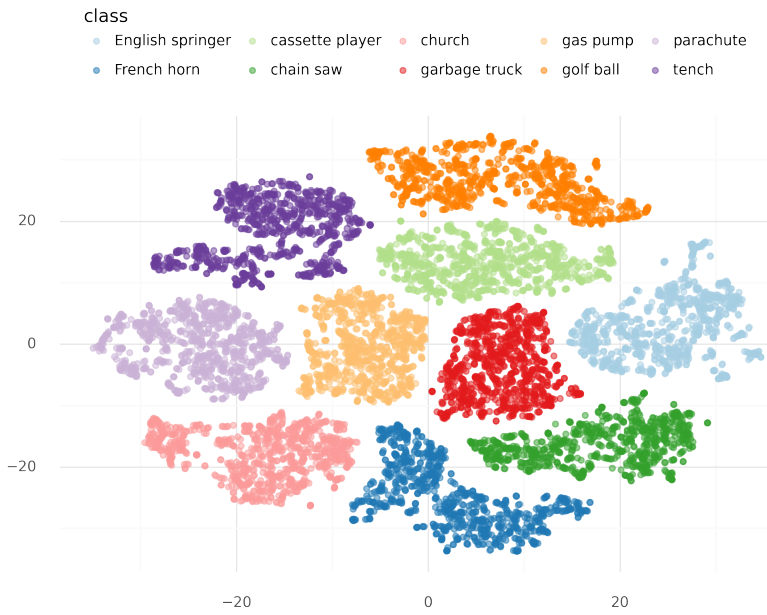


Figure 1: Yes, put a few words or sentences here explaining what is in the figure.

Figure 2 presents some summaries of the performance of our model. The left panel of Figure 2 presents something. The right panel of Figure 2 presents some other things.

6.3 Table Examples

Table 1 presents some summary of the data.

Table 2 presents some summaries of the performance of our model.

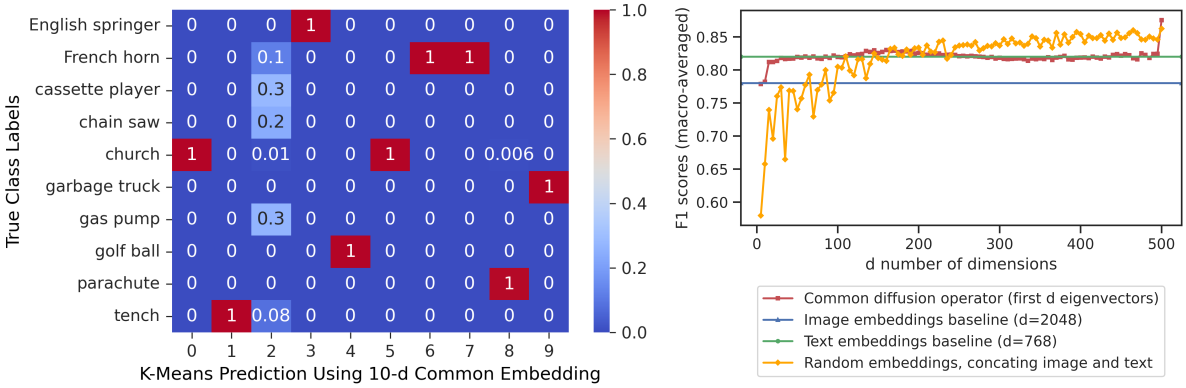


Figure 2: You can put figures side-by-side as well.

Table 1: Some Table Caption

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~100
Axon	Output terminal	~10
Soma	Cell body	up to 10^6

Table 2: Some Other Table Caption

Method	Modality	Edit distance	BLEU	METEOR	Precision	Recall	F1
PDF	All	0.255	65.8	82.1	77.1	81.4	79.2
GROBID	All	0.312	55.6	71.9	74.0	72.1	73.0
+ LaTeX OCR	Tables	0.626	25.1	64.5	61.4	80.7	69.7
	Plain text	0.363	57.4	69.2	82.1	70.5	75.9
	Math	0.727	0.3	5.0	11.0	8.6	9.7
Nougat base (350M*)	All	**0.071**	**89.1**	**93.0**	93.5	**92.8**	**93.1**
	Tables	0.211	69.7	79.1	75.4	80.7	78.0
	Plain text	0.058	91.2	94.6	96.2	95.3	95.7
	Math	0.128	56.9	75.4	76.5	76.6	76.5

6.4 Equations and Algorithms Examples

Algorithm 1 implements Fuzzy K-means.

Algorithm 1: Fuzzy K-means clustering algorithm

1. Choose primary centroids v_k
2. Compute the membership degree of all feature vectors in all clusters

$$u_{ki} = \frac{1}{\sum_{j=1}^{K} (\frac{D^2(x_i - v_k)}{D^2(x_i - v_j)})^{\frac{2}{m-1}}} \tag{1}$$

Algorithm 2 calculates net activation.

Algorithm 2: Computing Net Activation

Input: $x_1, \ldots, x_n, w_1, \ldots, w_n$
Output: y, the net activation
$y \leftarrow 0$;
for $i \leftarrow 1$ **to** n **do**
    $y \leftarrow y + w_i * x_i$;

In Variational Autoencoder (VAE), we directly maximize the Evidence Lower Bound (ELBO) using the following Equations 2–4.

$$\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{x},\boldsymbol{z})}{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\right] = \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p_\theta(\boldsymbol{x}\mid\boldsymbol{z})p(\boldsymbol{z})}{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\right] \tag{2}$$

$$= \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x}\mid\boldsymbol{z})] + \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log \frac{p(\boldsymbol{z})}{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\right] \tag{3}$$

$$= \underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x}\mid\boldsymbol{z})]}_{\text{reconstruction term}} - \underbrace{\mathscr{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}\mid\boldsymbol{x}) \,||\, p(\boldsymbol{z}))}_{\text{prior matching term}} \tag{4}$$

6.5 Inline Citation Examples

Citation in text (no parentheses): use `\cite{citekey}`. For example, Breiman (2001), Devlin et al. (2019).

Citation in parentheses: use `\citep{citekey}`. For example: (Vaswani et al. 2017), (Karras, Laine and Aila 2019).

To edit the contents of the "References" section, edit `reference.bib`. Many conference websites format citations in BibTeX that you can copy into `reference.bib` directly; you can also search for the paper on Google Scholar, click "Cite", and then click "BibTeX" (here's an example).

References

Breiman, Leo. 2001. "Statistical Modeling: The Two Cultures." *Statistical Science* 16(3): 199–215. [Link]

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics. [Link]

Karras, Tero, Samuli Laine, and Timo Aila. 2019. "A Style-Based Generator Architecture for Generative Adversarial Networks." In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. [Link]

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. "Attention Is All You Need." In *Advances in Neural Information Processing Systems*. [Link]

Appendices

A.1 Training Details A1
A.2 Additional Figures A1
A.3 Additional Tables A1

A.1 Training Details

A.2 Additional Figures

A.3 Additional Tables